DATA ANALYTICS REPORT

RETAIL DEMAND INTELLIGENCE: FORECASTING & DEMAND DRIVER ANALYSIS

Inventory Planning, Pricing Dynamics & Operational Performance

TOOLS USED

Python, Pandas, Numpy, Matplotlib, Plotly & ML libraries

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Overview & Business Objective

1.1 PROJECT OVERVIEW

Retail businesses operate in a dynamic environment where demand is influenced by factors such as historical sales, inventory, pricing, promotions, seasonality, and region.

Accurate demand estimation helps businesses maintain optimal inventory, reduce stockouts and excess stock, and improve procurement and operational planning.

The Retail Demand Intelligence project uses historical retail data and machine learning to predict demand and identify the key factors influencing it.

The project combines:

- Data understanding and cleaning
- Exploratory Data Analysis
- Feature engineering
- Machine learning regression
- Model evaluation and comparison
- Prediction-error analysis
- Model explainability
- Business-oriented insights

The objective is not only to predict demand accurately, but also to understand why the model makes its predictions & how these insights can support better decision-making.

1.2 PROBLEM STATEMENT

Retail demand is influenced by multiple interacting variables, making manual forecasting or reliance on historical averages insufficient for many situations.

The key problem addressed in this project is:

How can historical retail data be used to build a machine learning model that accurately predicts product demand while also identifying the major factors driving demand?

The project therefore treats Demand as the target variable and uses historical, operational, pricing, promotional, environmental, and temporal information as predictive features.

1.3 PROJECT OBJECTIVES

The major objectives of the project are:

- Develop a machine learning-based retail demand prediction system.
- Understand the underlying structure and characteristics of the dataset.
- Identify important patterns through exploratory data analysis.
- Create meaningful temporal, inventory, pricing, and historical features.
- Train and compare multiple regression models.
- Evaluate model performance using appropriate regression metrics.
- Investigate the largest prediction errors.
- Identify the most influential demand-driving features.
- Identify limitations and possible improvements for future development.

1.4 PROJECT SCOPE

The project follows a structured workflow covering data analysis, demand modeling, model explainability, and business interpretation to understand demand patterns, evaluate predictive performance, identify key demand drivers, and derive actionable retail insights.

01

Data Analysis

- Dataset inspection
- Data quality assessment
- Distribution analysis
- Relationship analysis

02

Demand Modeling

- Feature preparation
- Regression modeling
- Model comparison
- Model evaluation

03

Explainability

- Feature importance analysis
- Original-feature aggregation
- Investigation of large prediction errors

04

Business Interpretation

- Business Interpretation
- Inventory implications
- Pricing and promotion observations
- Seasonal and weather-related effects
- Recommendations for retail planning

Data Understanding & Exploratory Data Analysis

2.1 DATA OVERVIEW

The dataset provides a comprehensive view of retail operations across stores and products, covering inventory levels, sales activity, ordering patterns, pricing, discounts, promotions, weather conditions, seasonality, and demand. These variables provide the foundation for analyzing demand behavior, identifying operational patterns, and developing predictive models for retail demand.

Data Source: Kaggle

Dataset Size

76,000
Records

16
Original Features

Purchase Diversity

4
Regions

5
Stores

20
Products

DATA OVERVIEW

Below is a detailed description of the feature set:

Dataset Features	Type	Feature Description
Date	Date / Time	Date of the observation
Store ID	Categorical	Unique identifier of the store
Product ID	Categorical	Unique identifier of the product
Category	Categorical	Product category
Region	Categorical	Geographic region of the store
Inventory Level	Numerical (Discrete)	Inventory available at the time of observation
Units Sold	Numerical (Discrete)	Number of units sold
Units Ordered	Numerical (Discrete)	Number of units ordered
Price	Numerical (Continuous)	Selling price of the product
Discount	Numerical (Discrete)	Discount applied to the product
Weather Condition	Categorical	Weather condition during the observation
Promotion	Binary	Indicates whether a promotion was active
Competitor Pricing	Numerical (Continuous)	Price of the comparable product offered by competitors
Seasonality	Categorical	Seasonal classification of the observation
Epidemic	Binary	Indicates whether an epidemic-related condition was present
Demand	Numerical (Discrete)	Target variable representing product demand

2.2 TARGET VARIABLE

The target variable selected for prediction is:

Demand

Demand represents the quantity that the model attempts to estimate based on the available information

The target has:

104.32
Mean

100
Median

4
Minimum

430
Maximum

46.96
Standard Deviation

The distribution is concentrated around approximately 70-130 units, with a noticeable right tail extending toward higher demand values.

2.3 DATA QUALITY & PREPARATION OVERVIEW

Before modeling, the dataset was examined for:

- Data types
- Missing values
- Duplicate observations
- Categorical variables
- Numerical variables
- Temporal information
- Potential outliers
- Feature relationships

The *Date* variable was converted into usable temporal components, while categorical variables were prepared for machine learning using one-hot encoding.

2.4 Exploratory Data Analysis

Demand is concentrated around a stable baseline, with occasional high-demand spikes

$2,135
Average Cost per Service

Key observations

- Demand is concentrated primarily between approximately 70 and 130 units, with the highest frequency around 100 units.
- The distribution is right-skewed, with fewer observations extending to substantially higher demand values.

Business Insights

- Most retail activity occurs within a relatively predictable baseline demand range, which can assist with normal inventory planning.
- The long right tail indicates occasional high-demand events, highlighting the importance of maintaining sufficient safety stock and identifying demand spikes.

Groceries dominate the dataset, creating stronger representation of grocery demand patterns

$1,251

Typical Cost per Resource (Median)

Key observations

- Groceries has the largest representation in the dataset, with approximately 30,000 observations.
- Electronics has the smallest representation, while Furniture, Clothing, and Toys fall between these categories.

Business Insights

- Overall dataset patterns may be influenced more strongly by the highly represented Grocery category.
- Category-specific forecasting and inventory strategies can help account for differences in product behavior.

North dominates regional representation, while the remaining regions show balanced coverage

$2,147

Average Spending per Region

Key observations

- The North region has substantially more observations than the other regions.
- South, East, and West have relatively similar representation.

Business Insights

- Regional comparisons should account for the unequal number of observations.
- Regional segmentation can help retailers identify differences in demand behavior and plan inventory accordingly.

Retail demand fluctuates significantly over time, revealing recurring periods of elevated activity

Key observations

- Daily demand demonstrates considerable fluctuations throughout the observation period.
- Several recurring periods of lower and higher demand are visible, including elevated demand around parts of 2023.

Business Insights

- Retail demand should be treated as a time-dependent phenomenon rather than a constant value.
- Dynamic inventory and procurement planning can help businesses respond to expected periods of higher or lower demand.

Winter dominates seasonal representation, while the remaining seasons maintain relatively balanced coverage

Key observations

- Winter contains the largest number of observations.
- Spring, Summer, and Autumn have relatively similar representation.

Business Insights

- Seasonal representation should be considered when interpreting seasonal patterns.
- Incorporating seasonality into demand prediction allows the model to capture recurring changes associated with different periods of the year.

Demand is strongly linked to sales activity, while pricing factors show limited linear influence

52.03%

Typical CPU Utilization (Median)

Key observations

- Demand has a strong positive correlation with Units Sold (0.83).

- Demand has a moderate positive correlation with Units Ordered (0.51) and a weaker relationship with Inventory Level (0.13).

Business Insights

- Historical sales and ordering activity provide valuable information for understanding demand.
- Price and competitor pricing have very weak linear correlations with Demand in this dataset, suggesting that pricing effects may depend on interactions with other factors.

Important modeling note: The strong relationship between Demand and Units Sold is useful for exploratory analysis, but contemporaneous Units Sold should be interpreted carefully in a real forecasting scenario because future sales would not necessarily be known at prediction time.

Higher discounts do not consistently translate into higher demand, indicating a more complex pricing relationship

51.63%

Typical Memory Utilization (Median)

Key observations

- Demand varies substantially across every discount level.
- There is no obvious simple linear relationship where increasing discount consistently produces increasing demand.

Business Insights

- Larger discounts do not automatically guarantee proportionally higher demand.
- Discount strategies should therefore be evaluated alongside product characteristics, inventory, promotions, seasonality, and other demand drivers.

Sunny conditions are associated with higher average demand, while adverse weather corresponds to lower demand

Key observations

- Sunny conditions have the highest average demand, at approximately 115 units.
- Rainy and Snowy conditions have lower average demand, around 94-95 units.

Business Insights

- Weather conditions can contribute to differences in retail demand and should therefore be considered during forecasting.
- Businesses may benefit from adjusting inventory planning when weather patterns indicate potential changes in demand.

Data Preparation & Feature Engineering

3.1 DATA PREPARATION

Following the exploratory analysis, the dataset was prepared for machine learning.

The preparation process included:

- Selecting the target variable.
- Separating predictor variables from the target.
- Identifying numerical and categorical variables.
- Preparing categorical variables for encoding.
- Creating temporal features from the date.
- Preparing engineered inventory and pricing variables.
- Maintaining the chronological structure of the dataset.

3.2 TEMPORAL FEATURE ENGINEERING

The original ***Date*** variable was transformed into multiple temporal features:

- Year
- Month
- Week
- Day
- Day_of_Week
- Is_Weekend

These features allow the model to capture patterns associated with:

- Long-term yearly changes
- Monthly effects
- Weekly behavior
- Day-of-week effects
- Weekend vs weekday demand

3.3 INVENTORY FEATURES

Several inventory-related features were created to provide the model with additional operational information.

Inventory-to-Sales Ratio

- This feature represents the relationship between inventory availability and units sold.
- The resulting feature showed substantial variation, with a mean of approximately 4.28

Inventory Gap

- The inventory gap captures the difference between inventory and sales-related quantities.
- These features provide the model with additional information about the operational state of the retail system.

3.4 PRICING FEATURES

Pricing-related features included:

- Price_Difference
- Price_Difference_Percentage
- Promotion_Discount

These features were introduced to capture the relationship between the retailer's price, competitor pricing, and promotional activity.

3.5 HISTORICAL DEMAND FEATURES

Historical information was incorporated through:

- Previous_Demand
- Previous_Units_Sold
- Rolling_7_Day_Demand
- Rolling_7_Day_Sales

These features allow the model to incorporate recent demand and sales behavior rather than relying solely on current observations.

3.6 CATEGORICAL ENCODING

Categorical variables were transformed using ***OneHotEncoder***.

The categorical variables included:

- Store ID
- Product ID
- Category
- Region
- Weather Condition
- Seasonality

The preprocessing pipeline produced:

- 42 categorical features
- 22 numerical features
- 64 total model features

The use of a ***ColumnTransformer*** ensured that categorical and numerical variables were processed consistently within the modeling pipeline.

3.7 TRAIN-TEST SPLIT

The dataset was divided chronologically into training and testing portions.

The split used:

- Training observations: 63,800
- Testing observations: 12,200

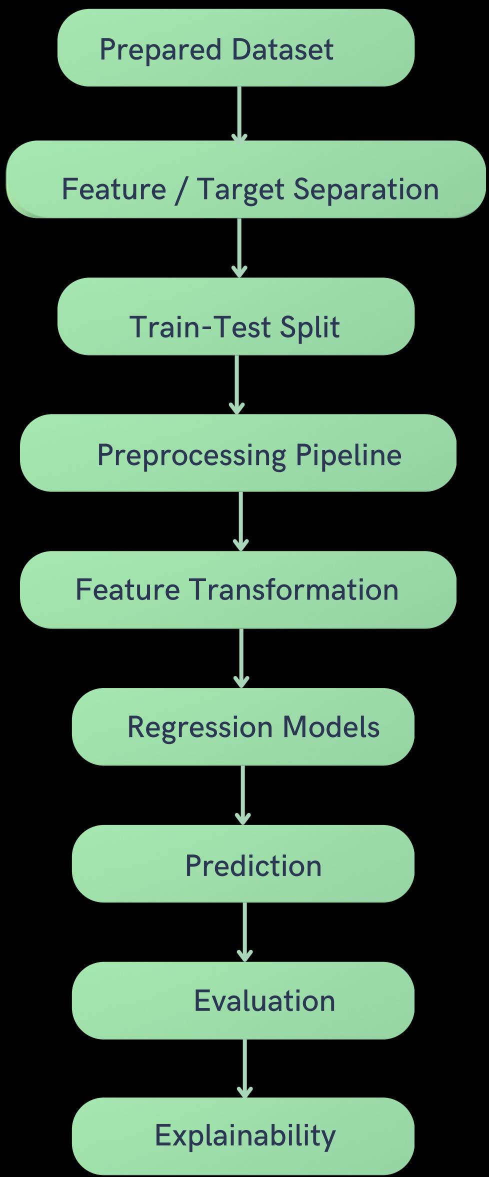
A chronological split was used rather than a random split to better preserve the temporal structure of the retail demand problem.

Model Development & Experimentation

4.1 MODELING APPROACH

The problem was formulated as a supervised regression problem, with ***Demand*** as the continuous target variable.

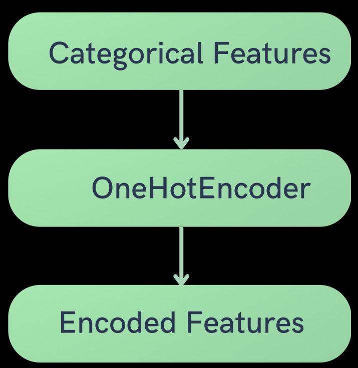
The modeling workflow consisted of:



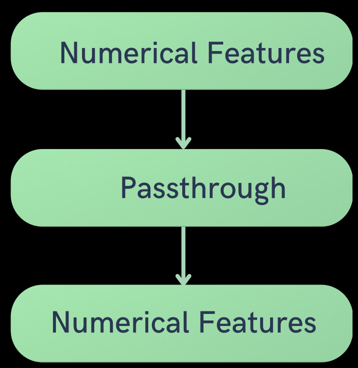
4.2 PREPROCESSING PIPELINE

A unified preprocessing pipeline was used to ensure that the same transformations were applied consistently during training and prediction.

Demand Modeling



Numerical Pipeline



Both were combined through a *ColumnTransformer*.

4.3 REGRESSION MODELING

Multiple regression approaches were considered rather than relying on a single algorithm.

The objective was to identify a model capable of:

- Capturing nonlinear relationships
- Handling mixed feature types
- Learning interactions between variables
- Producing accurate demand predictions

The models were evaluated using consistent training and testing data.

Model	MAE	RMSE	R^2
Baseline	36.491773	45.419423	-0.021845
Linear Regression	15.154549	20.365633	0.794554
Ridge Regression	15.154426	20.365599	0.794555
Lasso Regression	15.878328	21.387819	0.773413
Decision Tree Regression	17.284016	23.607680	0.723937

DATA OVERVIEW

Below is a detailed description of the feature set:

Model	MAE	RMSE	R^2
Random Forest Regression	12.218563	16.094710	0.871688
Gradient Boosting Regression	13.429953	17.480096	0.848648
XGBoost Regression	11.580912	15.131741	0.886583
Extra Trees Regression	10.876563	14.656385	0.893597
HistGradientBoosting Regression	11.783125	15.291003	0.884183
Voting Regression	11.044994	14.545320	0.895203
Stacking Regression	10.750887	14.194631	0.900196
Tuned Extra Trees Regression	11.018017	14.809778	0.891358
Tuned XGBoost Regression	11.402505	14.890003	0.890177
Tuned Random Forest Regression	12.158307	16.001342	0.873172
Final Stacking Regression	10.628791	14.016545	0.902684

Model Evaluation & Comparison

5.1 EVALUATION METRICS

The model was evaluated using three primary regression metrics.

Mean Absolute Error (MAE)

- MAE measures the average absolute difference between actual and predicted demand.
- A lower MAE indicates that predictions are closer to actual demand on average.

Root Mean Squared Error (RSME)

- RMSE gives greater weight to larger prediction errors.
- This makes it particularly useful for identifying models that occasionally produce large forecasting mistakes.

R^2 Score

- R^2 measures the proportion of variation in the target that is explained by the model.
- A higher R^2 indicates stronger explanatory performance.

5.2 ERROR ANALYSIS

Beyond overall metrics, prediction errors were examined at multiple levels.

This included analysis by:

- Category
- Region
- Demand Level
- Month

This was important because an overall metric can hide areas where a model performs particularly well or poorly.

5.2.1 ERROR BY CATEGORY

Results obtained:

Category	MAE	RMSE
Groceries	12.62	16.23
Clothing	11.69	15.20
Electronics	10.06	12.78
Toys	9.24	12.25
Furniture	6.71	8.64

Key observations

- Groceries has the highest MAE and RMSE among the categories.
- Furniture has the lowest prediction error.

This indicates that model performance varies depending on product category.

5.2.2 ERROR BY REGION

Results obtained:

Region	MAE	RMSE
East	10.92	14.50
North	10.34	13.80
South	11.20	14.34
West	10.35	13.61

Key observations

- South has the highest MAE.
- West has the lowest RMSE.
- Overall regional differences are relatively moderate.

This suggests that the model performs reasonably consistently across regions, although some regional variation remains.

5.2.3 ERROR BY DEMAND LEVEL

Results obtained:

Demand Level	MAE	RMSE
Very Low	7.07	9.39
Low	9.54	12.23
Medium	9.89	12.34
High	11.33	14.14
Very High	15.39	19.86

Key observations

- The error increases substantially as demand increases.

The Very High demand group has the largest prediction error, indicating that the model finds demand spikes more difficult to predict accurately.

(This is one of the most important findings from the evaluation stage)

5.2.4 ERROR BY MONTH

Results obtained:

Month	MAE	RMSE
2023-10	9.94	13.26
2023-11	10.63	13.80
2023-12	11.09	14.60
2024-01	10.86	14.37

Key observations

- Prediction error varies across the testing months, with December showing the highest MAE and RMSE among the evaluated months.

This suggests that certain periods may introduce additional demand variability that is more difficult for the model to capture.

5.3 LARGEST PREDICTION ERRORS

The largest prediction errors were specifically investigated rather than treating them as random noise.

The largest observed error included:

- Date: 2023-10-26
- Store: S005
- Product: P0015
- Category: Groceries
- Region: North
- Actual Demand: 237
- Predicted Demand: 153.77
- Absolute Error: 83.23

The model significantly underestimated demand for this observation.

5.4 INVESTIGATION OF THE LARGEST ERROR

Relevant information for the observation included:

Feature	Value
Inventory Level	58
Units Sold	58
Units Ordered	380
Discount	25
Promotion	1
Competitor Pricing	49.30
Price	57.76
Previous Demand	73
Rolling 7-Day Demand	74.86
Actual Demand	237
Predicted Demand	153.77

The model significantly underestimated demand for this observation.

5.4 INVESTIGATION OF THE LARGEST ERROR

The model had evidence of elevated activity through the historical demand features, but the actual demand was substantially higher than the recent historical pattern.

Explainability conclusion

The model recognized the demand increase but failed to fully capture an unusually large demand spike.

(This is valuable because it identifies a specific model weakness rather than simply reporting an R^2 score)

Explainability & Final Model

6.1 WHY EXPLAINABILITY WAS INCLUDED

Accurate predictions alone are not always sufficient for business applications.

Retail decision-makers also need to understand:

- Which variables influence predictions?
- Which operational factors matter most?
- What drives model behavior?
- Where does the model struggle?

Feature importance analysis was therefore performed to understand the model's decision-making behavior.

BUSINESS / DEVELOPER TAKEAWAYS

- **Network Architecture Is a Primary Cost Lever:** Cloud spend is driven more by traffic routing and data transfer patterns than raw compute usage, making network design and egress optimization the highest-impact cost reduction opportunity.

- **Cost Is Highly Concentrated in a Small Resource Subset:** A limited number of high-cost Resource IDs disproportionately influence total expenditure, making targeted deep-dive optimization far more effective than broad cost-cutting efforts.

- **Overprovisioning Is Inflating Infrastructure Spend:** Costs are driven more by allocated instance size and configuration than actual CPU or memory utilization, indicating significant rightsizing and autoscaling opportunities.
- **Resource Configuration Drives More Cost Than Runtime Duration:** High-spec instance types and premium configurations materially increase spend even for shorter workloads, making configuration review more impactful than simply reducing runtime.

- U**tilization Imbalances Signal Immediate Efficiency Gains:** Imbalanced CPU–memory usage and sub-50% utilization across high-cost services highlight clear opportunities for workload tuning, resizing, and instance optimization.